Common Waveform Estimation (CWE) toolbox ver.3

Updated on 2010/11/30 by Yusuke Takeda

Common Waveform Estimation (CWE) toolbox is a suite of MATLAB functions to execute a common waveform estimation method developed by Takeda et al. (2010). From single/multi-channel EEG/MEG epochs, these functions estimate the number of waveforms common across trials, their delays in individual trials, and all the waveforms.

※ Hereafter temporal patterns of EEGs/MEGs common across trials are referred to as waveforms.

1. Overview of the waveform estimation method

The method consists of three steps: Delay estimation, Waveform estimation, and Evaluation, which are consecutively repeated as shown in Fig. 1. At the beginning of the procedure, the number of waveforms is set to 1. In the Delay estimation step, delays of waveforms are estimated using preset number of waveforms. In the Waveform estimation step, waveforms are estimated using the estimated delays and the preset number of waveforms. In the Evaluation step, the preset number of waveforms is evaluated using the estimated waveforms and delays. If the preset number of waveforms is evaluated as wrong, we return to the Delay estimation step by increasing preset number of waveforms by 1. By repeating this procedure until preset number of waveforms is evaluated as true, the number of waveforms, their delays, and all the waveforms are simultaneously estimated.

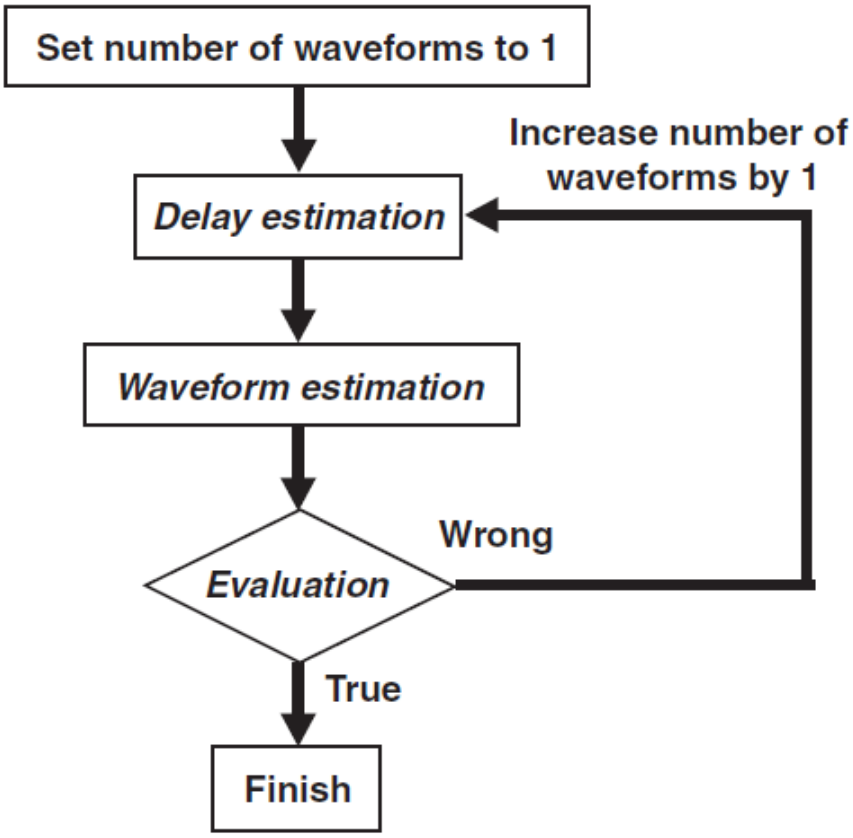


Figure 1: Flowchart of the waveform estimation method.

2. Installation

Download and unzip the file "CWE.zip" wherever you like. Please start from demo functions "demo_*.m" to learn how the functions work.

-Example
>>**demo_simulated_data**

※ Note that a function "evaluate.m" needs a statistics toolbox.

3. Main functions

There are three classes of functions: (1) demo functions, (2) function for overall procedure,

and (3) functions for individual steps (see Fig. 2).

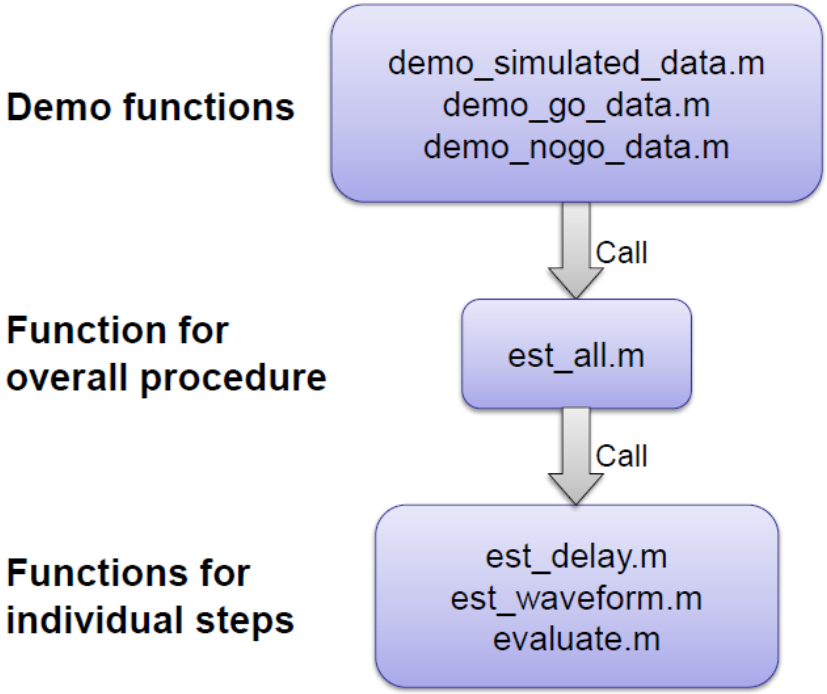


Figure 2: Main functions.

(1) Demo functions

These are for demonstration purpose. You can learn how functions in CWE toolbox work using some examples. Please run these functions at first.

・**demo_simulated_data.m** : a demo function using simulated data.

・**demo_go_data.m** : a demo function using real EEG data during Go trials of a Go/NoGo task.

・**demo_nogo_data.m** : a demo function using real EEG data during NoGo trials of a Go/NoGo task.

(2) Function for overall procedure

This executes the overall waveform estimation method (see Fig. 1).

・**est_all.m** : this estimates the number of waveforms common across trials, their delays, and all the waveforms from single/multi-channel EEG/MEG epochs.

(3) Functions for individual steps

These are for individual steps.

・**est_delay.m** : this estimates the delays of waveforms. This is for "Delay estimation" step.

・**est_waveform.m** : this estimates all the waveforms. This is for "Waveform estimation"

step.

・**evaluate.m** : this evaluates whether a preset number of waveforms is true. This is for "Evaluation" step. A statistics toolbox is needed.

4. Examples of usages

Examples of usages of CWE toolbox are demonstrated in demo functions (demo_*.m). Below, other usages of the individual functions are described. Suppose that a variable "EEG_epoch" (Ntime x Ntrial x Nchannel) contains EEG data that are triggered on stimulus onsets.

(1) Average EEG/MEG epochs

When the number of waveforms is one, "Waveform estimation" step corresponds to the averaging procedure. For example,

>>**stimulus_triggered_average = est_waveform(EEG_epoch, zeros(Ntrial,1));**

returns stimulus-triggered average EEGs.

>>**response_triggered_average = est_waveform(EEG_epoch, reaction_time);**

returns response-triggered average EEGs, where "reaction_time" (Ntrial x 1) contains time points of response onsets.

(2) Decomposition of EEG/MEG into stimulus- and response-locked components

In EEGs/MEGs during reaction time tasks, stimulus- and response-locked EEG/MEG components exist and these are sometimes temporally overlapping. In such cases, stimulus- and response-triggered averaging procedures cannot extract exact waveforms of the components because these are mutually contaminated by the averaging procedure. "Waveform estimation" step extracts pure waveforms that are not contaminated with each other. For example,

>>**delay = [zeros(Ntrial,1), reaction_time];**

>>**waveform = est_waveform(EEG_epoch, delay);**

return the pure waveforms of stimulus- and response-locked components.

(3) Estimate waveforms of variable-delay components

In EEGs/MEGs during covert response tasks, such as decision-making tasks, the delays of waveforms related to covert responses are variable and unknown. In such cases, the averaging procedure cannot be used. "Delay estimation" step can estimate such variable and unknown delays. For example,

>>**delay = est_delay(EEG_epoch, N);**
returns N sets of delays (Ntrial x N), where "N" indicates the number of waveforms common across trials. Once delays are estimated, N waveforms can be estimated by
>>**waveform = est_waveform(EEG_epoch, delay);**

5. Sample EEG data

CWE toolbox contains sample EEG data during a Go/NoGo task ("go_data.mat" & "nogo_data.mat"). Detailed information of the data is described below.

(1) Variables in "go_data.mat"
- **data** : EEG data during Go trials [Ntime (200) x Ntrial (96) x Nchannel (5)].
- **reaction_time** : reaction times (ms) of Go trials [Ntrial (96) x 1].
- **time_label** : time labels (ms) of the variable "data" [Ntime (200) x 1].
- **channel_label** : channel labels of the variable "data".

(2) Variables in "nogo_data.mat"
- **data** : EEG data during NoGo trials [Ntime (200) x Ntrial (100) x Nchannel (5)].
- **time_label** : time labels (ms) of the variable "data" [Ntime (200) x 1].
- **channel_label** : channel labels of the variable "data".

(3) Experimental procedure
A subject participated in four experimental blocks, each consisting of 50 trials. He was instructed to push a button immediately after a "Go" signal (green LED) and to not push it after a "NoGo" signal (red LED). The green and red LEDs were illuminated in random order at almost equal probability. In two blocks, he had to respond with their right index finger and in the other two blocks with their left index finger. In off-line analysis, the data from the blocks of the left and right fingers were mixed.

(4) EEG recording
During the task, surface EEGs were recorded from 19-ch tin electrodes, mounted in a cap (Electro-Cap International, Inc., Eaton, Ohio, USA) based on the international 10-20 system. A tin electrode placed on AFz was used as a ground. The EEGs were amplified on a Nihon Kohden EEG-1100 with a time constant of 0.3 s. Ag/AgCl electrodes were placed on both earlobes and recorded their potentials separately. Their averaged potentials were subtracted from the EEGs off-line. The sampling rate was 1000 Hz.

(5) Off-line analysis

The EEGs at Fz, C3, C4, Cz, and Pz were selected. The EEGs were resampled at a rate of 100 Hz. They were filtered with a bandpass of 2—40 Hz using two kinds of finite impulse response (FIR) filters: a high-pass filter of 2 Hz (300-point, -26 dB at 1 Hz) and a low-pass filter of 40 Hz (15-point, -45 dB at 50 Hz). Then the filtered EEGs were segmented into 2 s epochs from -500 to 1500 ms after stimulus onsets.

The reaction times of the Go trials were defined as the intervals between the stimulus and the button-push signal (response) onsets. The Go trials with an RT either shorter than 100 ms or longer than 400 ms and the NoGo trials with any response were excluded. An artifact criterion of -50 μV was used for the EEGs to reject trials with excess ocular artifacts or measurement noise. At last 96 Go trials and 100 NoGo trials were obtained.

6. Referencing the toolbox

When using this tool for a paper please refer to the following paper:
Takeda Y., Sato M., Yamanaka K., Nozaki D., Yamamoto Y., A generalized method to estimate waveforms common across trials from EEGs. NeuroImage 51 (2010) 629-641.

7. Feedback & Bug report

Any feedback and bug report are welcome. Please keep contact with me (takeda@atr.jp). I would like to respond as quickly as possible.

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9. Acknowledgements

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